Statistical Techniques for Social Research

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Bachelor's Degree in Political Science (2024/2025) – University of Barcelona

The course is split into three sections following the three possible research objectives: effect estimation, description, and prediction.

Schedule

Week 1

- Tuesday: Reading of the syllabus and *Political Science as a Vocation* (Keohane, 2009) [2 hours]
- Wednesday: Lecture 1 (Starting line) [2 hours]
- Thursday: Reading Chapter 1 of *Designing Social Inquiry* (King, Keohane, and Verba, 1994) [2 hours]
- Friday: Lecture 2 (Science for researching politics and policies) [2 hours]

Week 2

- Tuesday: Reading Chapter 2 of Llaudet & Imai (2023) [2 hours]
- Wednesday: Lecture 3 (Estimating effects with experimental data) [2 hours]
- Thursday: Complete Exercise Sheet 1 [2 hours]
- Friday: Lecture 4 (Estimating effects with observational data) [2 hours]

Week 3

- Tuesday: Complete Exercise Sheet 2 [2 hours]
- Wednesday: Lecture 5 (Estimating effects with observational data II and in-class exercises with SPSS) [2 hours]
- Thursday: Read Sections 3 to 3.4.3 of Llaudet & Imai (20023) [2 hours]
- Friday: Lecture 6 (More exercises on the scientific research process and effect estimation) [2 hours]

Week 4

- Tuesday: Reading half of Chapter 5 of Gerring (2014) [2 hours]
- Wednesday: Lecture 7 (Describing a political phenomenon: sampling and central tendency parameters I) [2 hours]
- Thursday: Reading the other half of Chapter 5 of Gerring (2014) [2 hours]
- Friday: Lecture 8 (Describing a political phenomenon: sampling and central tendency parameters II) [2 hours]

Week 5

- Tuesday: Study readings and lecture notes [2 hours]
- Wednesday: Lecture 9 (Describing a political phenomenon: other parameters of interest I) [2 hours]
- Thursday: SPSS exercises on description [2 hours]
- Friday: Lecture 10 (Describing a political phenomenon: other parameters of interest II and general review) [2 hours]

Week 6

- Monday: Study readings and lecture notes [2 hours]
- Tuesday: Study readings and lecture notes [2 hours]
- Wednesday: First midterm exam [2 hours]
- Thursday: Reading *Mere Description* (Gerring, 2012) [2 hours]
- Friday: Lecture 11 (Describing a political phenomenon with two categorical variables) [2 hours]

Week 7

- Tuesday: Reading Sections 3.5 and 3.6 of Llaudet & Imai (2023) [2 hours]
- Wednesday: Lecture 12 (Describing a political phenomenon with two numerical variables) [2 hours]
- Thursday: SPSS exercises on description with two variables [2 hours]
- Friday: Lecture 13 (In-class exercises on description with two variables) [2 hours]

Week 8

- Tuesday: Review notes and readings on all association parameters [2 hours]
- Wednesday: Lecture 14 (More exercises on description with two variables) [2 hours]
- Thursday: Read Sections 4 to 4.3.4 of Llaudet & Imai (2023) [2 hours]
- Friday: Lecture 15 (Predicting a political phenomenon: introduction to the regression model) [2 hours]

Week 9

- Tuesday: Review mathematical concepts: summation, functions, and derivatives with *Math Prefresher for Political Scientists* (Signorino et al., 2024) [2 hours]
- Wednesday: Lecture 16 (Predicting a political phenomenon: ordinary least squares estimation) [2 hours]
- Thursday: Review mathematical concepts: derivatives and optimisation with class notes and *Math Prefresher for Political Scientists* (Signorino et al., 2024) [2 hours]
- Friday: Lecture 17 (Predicting a political phenomenon: model evaluation parameters) [2 hours]

Week 10

- Wednesday: Review Week 9 lectures and read Sections 4.4 to 4.7 of Llaudet & Imai (2023) [2 hours]
- Thursday: Complete Exercise Sheet 5 [2 hours]

Week 11

- Wednesday: Complete Exercise Sheet 6 in groups [2 hours]
- Friday: Continue Exercise Sheet 6 in groups and prepare the presentation [2 hours]

Week 12

- Tuesday: Reading Chapter 4 of Imai (2018) [2 hours]
- Wednesday: Lecture 18 (Correction of Exercise Sheet 5) [2 hours]
- Thursday: Exercise on outliers in a regression model [2 hours]
- Friday: Review corrections of Exercise Sheet 6 and resubmit presentation [2 hours]

Week 13

- Tuesday: Reading Chapter 4 (up to Section 4.3 and Appendix 4.2, inclusive) of Stock & Watson (2012) [2 hours]
- Wednesday: Lecture 19 (Presentation of Exercise Sheet 6) [2 hours]
- Thursday: Study/review materials on bivariate association parameters: Pearson's chi-square, Cramer's V; Pearson correlation and Spearman correlation [2 hours]
- Friday: Complete Exercise Sheet 7 [2 hours]

Week 14

- Tuesday: Study material on bivariate description and prediction [2 hours]
- Wednesday: Lecture 20 (Introduction to multivariable analysis and influential outliers) [2 hours]
- Thursday: Study material on bivariate description and prediction [2 hours]
- Friday: Lecture 21 (Review of bivariate description and prediction techniques) [2 hours]

Week 15

- Monday: Study material on bivariate description and prediction [2 hours]
- Tuesday: Study material on bivariate description and prediction [2 hours]
- Wednesday: Second midterm exam [2 hours]
- Thursday: Reading IPC methodological document (INE, 2017) [2 hours]
- Friday: Lecture 22 (Bivariable description of time series: index numbers) [2 hours]

Week 16

- Monday: Lecture 23 (Review of the entire course) [2 hours]
- Tuesday: Study the entire course [2 hours]
- Wednesday: Lecture 24 (Review of the entire course) [2 hours]
- Thursday: Study the entire course [2 hours]

Exercise sheets

1. Is there racial discrimination in the labour market? [Based on Llaudet]
2. Does negative political advertising on television reduce voter turnout? [Based on Llaudet]
3. The vote for Vox in the 2024 Catalan Parliament elections
4. Describing the level of development of different countries according to different dimensions
5. Prediction of votes in favour of Obama based on bets on 'Obama will win' [Based on Imai (2018)]
6. Predicting Hungary's level of democracy
7. Bivariate association parameters versus regression models

References

Llaudet, E., & Imai, K. (2023). *Data analysis for social science: A friendly and practical introduction*. Princeton University Press. [Main textbook]

Gerring, J. (2014). *Metodología de las ciencias sociales: un marco unificado*. Alianza.

Gerring, J. (2012). Mere description. *British Journal of Political Science*, *42*(4), 721–746.

Imai, K. (2018). *Quantitative social science: An introduction*. Princeton University Press.

INE (2017). Índice de Precios de Consumo. Base 2016. Metodología. https://www.ine.es/metodologia/t25/t2530138_16.pdf

King, G., Keohane, R. O., & Verba, S. (1994). *Designing social inquiry: Scientific inference in qualitative research*. Princeton University Press.

Keohane, R. O. (2009). Political science as a vocation. *PS: Political Science & Politics*, *42*(2), 359-363.

Signorino, R. T., Scheve, K., Dickson, E., Kedar, O., Fowler, J., Imai, K., Kline, J., Epstein, D., Ansell, B., Moore, R., Kellermann, M., Powell, E., Katkin, J., Lam, P., Rios, V., Pan, J., Kashin, K., Prillaman, S., Pettigrew, S., *et al.* (2024). *Math Prefresher for Political Scientists*. [Online publication]. https://iqss.github.io/prefresher/

Stock, J. H. & Watson, M. W. (2012). *Introducción a la Econometría*. Pearson.